METRO INTERSTATE TRAFFIC ANALYTICS

Part 2 | Methodology and findings

Purpose. This report summarizes the four implemented Python modules in Part-2: a raw-data cleaning pipeline, feature engineering, four descriptive visualizations, and a command-line query app. The work is descriptive and prepares data for possible future modelling; no predictive model is trained or evaluated.

Dataset and workflow

The supplied source contains 48,204 hourly records. The processed dataset contains 40,575 unique timestamps spanning 2 Oct 2012-30 Sep 2018. Coverage is uneven and includes missing months; hourly records are observations, not a complete continuous time series.

Methodology

Stage	Implemented method and output
1 Clean	pipeline.py validates the nine required columns and day-first timestamps, standardizes category text, removes 17 exact duplicate rows and 7,612 additional duplicate timestamps (first row kept), then writes 40,575 records. Ten non-positive temperatures and one rainfall value above 9,000 mm are replaced by the corresponding calendar-month median. The known 23-24 Jul 2016 low-traffic outage (48 hourly records) is retained and warned about.
2 Engineer	feature_engineering.py retains the input columns and adds hour, weekday, weekend, cyclical-hour sine/cosine, rain/snow indicators, full-dataset z-scores for temperature/rainfall, one-hot weather fields, traffic category, and congestion flag. Outputs are a 63-column feature CSV; thresholds are Low <4,500, Medium 4,500-5,500 inclusive, High >5,500 vehicles/hour.
3 Visualize	visualize.py creates hourly mean demand, weekday/weekend hourly profiles, mean traffic by every weather category with sample counts, and monthly congestion rate with record coverage. Missing months break the trend line; July 2016 is marked for the known outage. Each successful PNG save is logged with its path.
4 Query	traffic_app.py provides date/hour, top-N peak, weekday/weekend compare, quietest-hour recommendation, and date-specific weather commands. It loads the cleaned CSV once per invocation, validates dates/hours, formats tables, and reports errors without raw tracebacks. Recommendations rank pooled historical hourly means.

Definitions

Congestion is defined as traffic_volume > 5,500. This is a project definition, not a threshold estimated from this application. Day of week is derived from the parsed timestamp; weekdays are Monday-Friday and weekends Saturday-Sunday.

OBSERVED RESULTS

Descriptive findings from the cleaned hourly records

Key findings

Pattern	Observed result
Daily demand cycle	Mean traffic peaks at 16:00 (5,708.6 vehicles/hour) and bottoms at 03:00 (373.2). Hour-of-day averages reveal a pronounced overnight/daytime pattern.
Weekday vs weekend	Weekday mean: 3,557.4 vehicles/hour across 28,979 records. Weekend mean: 2,623.9 across 11,596. These are descriptive differences, not causal estimates.
Congestion distribution	6,107 of 40,575 hours (15.05%) exceed 5,500. Categories: Low 26,060 (64.2%); Medium 8,408 (20.7%); High 6,107 (15.1%).
Weather association	Clouds: mean 3,617.0 (n=15,123); Clear: 3,055.3 (n=13,366); Rain: 3,355.3 (n=4,769). Weather groups have different time-of-day mixes, so these gaps do not establish weather effects.
Time coverage	There are records in 63 of 72 calendar months between the first and last observations; Sep 2014-May 2015 is entirely absent. The chart exposes the gap rather than interpolating it.

Interpretation and limitations

- The strongest measured pattern is temporal: average traffic changes substantially by hour and differs between weekdays and weekends. All summaries describe the available sample only.
- Weather comparisons are unadjusted associations. Cloudy/hazy observations are more concentrated in daytime, and rare groups have little support (Squall n=1, Smoke n=15, Fog n=201, Thunderstorm n=453); interpret their means cautiously.
- Uneven coverage, missing months, and the preserved July 2016 outage limit trend interpretation. Monthly congestion rates use observed records only; they are not imputed to missing hours.
- Feature z-scores are calculated over the complete dataset. For future time-based model evaluation, fit scaling and encoders on training data only to avoid information leakage. The current feature script does not spread the holiday flag across every hour of its date.
- The quietest hours CLI result is a pooled historical ranking, not a date-specific forecast or a travel guarantee. No model accuracy or causal impact has been assessed.

Reproducible outputs

Run from the repository root: python Part-2/pipeline.py - python Part-2/feature_engineering.py - python Part-2/visualize.py; then query with python Part-2/traffic_app.py --help. Outputs are the cleaned and feature CSVs, four PNG figures under Part-2/outputs/visualizations/, and per-script logs under Part-2/logs/.

Evidence base: Part-2 source modules and the generated cleaned/feature datasets, visualization outputs, and pipeline/app logs. Findings are descriptive; no unverified forecasts or causal claims are made.